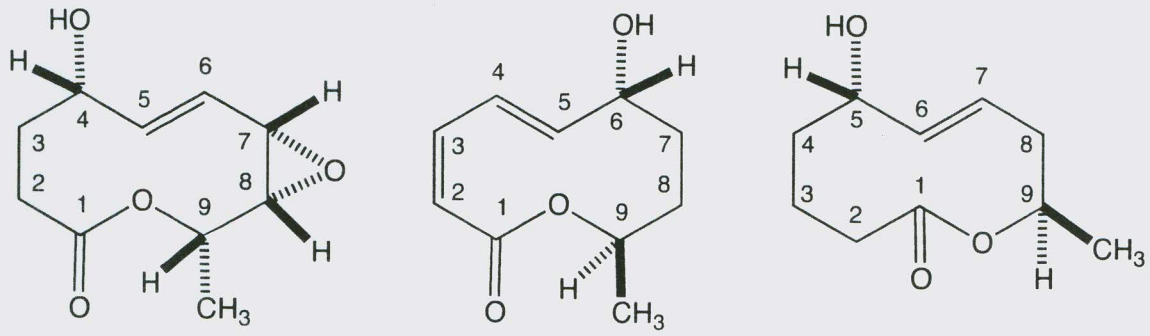
Organic Cumulative Exam Dec 1, 2011

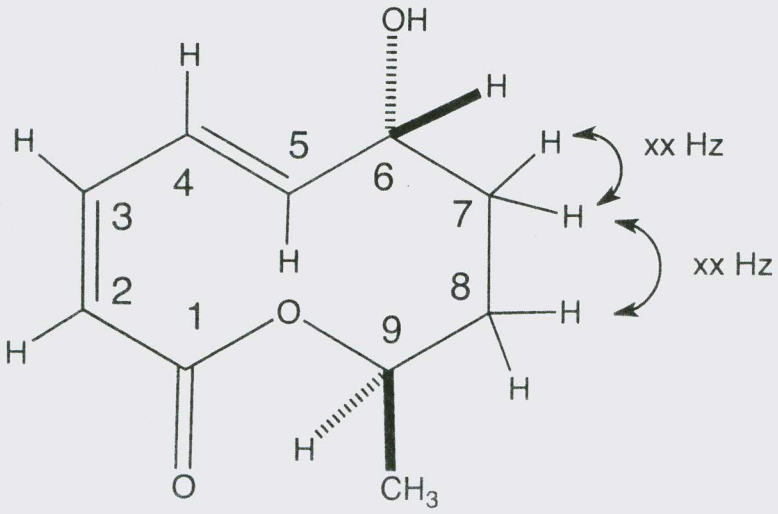
The following compounds were isolated from the fungus *Stagonospora cirsii* which grows on the noxious weed *Cirsium arvense*, and is being considered as a potential natural herbicide for the weed (*J. Nat. Prod.* **2008**, *71*, 31-34). On the following page are three ^{1}H and ^{13}C NMR data sets that were obtained for the structures.

Q.1. Match the data sets to the structures. **Briefly** indicate what data you used to match each structure to each data set. *You do not need to interpret all of the data for this question, just enough to distinguish one from the others.*



Stagonolide D Stagonolide E Stagonolide F

Q.2 For the data set you identified for Stagonolide E, assign all of the proton and carbon resonances. Indicate coupling constants between protons with arrows as shown below, and multiplicities of carbons to support your assignments.



Stagonolide E

Data set #1:
6.60 (br d, J=11.6 Hz, 1H), 6.12 (br d, J=15.4 Hz, 1H), 5.84 (d, J=11.6 Hz, 1H), 5.73 (dd, J=15.4, 9.6 Hz, 1H), 4.98 (dq, J=11.7, 6.5 Hz, 1H), 2.08 (ddd, J=14.2, 9.0, 3.8), 1.85 (dd, J=15.8, 9.0 Hz, 1H), 1.73 (ddd, J=14.2, 9.5, 9.0 Hz, 1H), 1.21 (d, J=6.5 Hz, 3H).

21.4 (CH3), 30.4 (CH2), 37.4 (CH2), 73.2 (CH), 73.7 (CH), 125.6 (CH), 126.6 (CH), 139.6 (CH), 140.2 (CH), 168.2 (C).

Data set #2:
5.64 (dd, J=17.0, 4.8 Hz, 1H), 5.52 (dd, J=17.0, 8.4, 1H), 5.34 (dq, J=6.7, 2.6 Hz, 1H), 4.13 (ddd, J=8.4, 7.4, 4.3 Hz, 1H), 3.65 (dd, J=4.8, 3.9 Hz, 1H), 3.05 (dd, J=3.9, 2.6 Hz, 1H), 2.28 (ddd, J=13.0, 7.4, 1.8 Hz, 1H), 2.11 (dd, J=13.0, 11.2 Hz, 1H), 2.05 (dd, J=14.0, 1.8 Hz, 1H), 2.00 (ddd, J=14.0, 11.2, 4.3 Hz, 1H), 1.37 (d, J=6.7 Hz, 3H).

16.2 (CH3), 31.2 (CH2), 35.0 (CH2), 55.4 (CH), 58.2 (CH), 65.7 (CH), 75.1 (CH), 128.1 (CH), 134.2 (CH), 173.5 (C).

Data set #3
5.66 (ddd, J=15.1, 10.7, 3.4 Hz, 1H), 5.27 (br dd, J=15.1, 10.2 Hz, 1H), 5.15 (ddq, J=12.7, 6.4, 3.4 Hz, 1H), 4.06 (ddd, J=10.2, 9.3, 3.3 Hz, 1H), 2.35 (m, 1H), 2.30 (m, 1H), 2.05 (m 2H), 2.00 (m, 3H), 1.18 (d, 6.4 Hz, 1H).

21.7 (CH3), 31.5 (CH2), 32.1 (CH2), 34.3 (CH2), 35.0 (CH2), 75.4 (CH), 131.3 (CH), 134.5 (CH), 174.8 (C)